

Event-Based Dynamic Turbulence Mitigation

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Abstract—Atmospheric turbulence induces coupled spatio-temporal distortions, including blur, geometric deformation, and temporal jitter, which severely degrade image quality. We propose EvTurM, a practical framework leveraging event camera data for dynamic turbulence mitigation with precise motion cues and stable temporal modeling. Leveraging the high temporal resolution and dynamic range of events, EvTurM achieves robust restoration under diverse turbulence conditions. EvTurM comprises two key modules: (1) the event-aware modality enhancement module, which uses event-derived motion to enrich RGB features and recover structural details, and (2) the bidirectional modality calibration module, which jointly aligns RGB and event features in forward and backward propagation to reduce misalignment and enhance temporal consistency. Extensive experiments show EvTurM consistently surpasses existing methods and achieves superior performance.

Index Terms—Atmospheric turbulence, event camera, video restoration, multi-modal fusion.

I. INTRODUCTION

ATMOSPHERIC turbulence is a common phenomenon in long-range imaging systems, such as ground surveillance, satellite remote sensing, and aerial reconnaissance. It arises from spatiotemporal fluctuations in air refractive index, producing space-variant blur, nonrigid geometric warps, and inter-frame jitter. Similar blur phenomena have been widely studied in the image restoration and deblurring literature [9], [29], [30], [31], although turbulence-induced distortions are complicated by dynamic refractive effects. These effects jointly degrade spatial fidelity and temporal coherence, and their tight coupling makes restoration challenging.

To address turbulence-induced degradation, existing methods fall into single-frame and multi-frame approaches. Single-frame methods [6], [15], [16] recover a clean image from one observation, but long exposures integrate time-varying, anisoplanatic warps into a non-linear, space-variant operator (not a simple convolution), so high-frequency detail is irreversibly lost and networks over-smooth or hallucinate; the lack of temporal cues further limits robustness under strong dynamics. Multi-frame methods [8], [32], [33] exploit temporal redundancy

via alignment and fusion, yet real turbulence induces spatially varying, rapidly changing misalignments and scintillation that breaks brightness constancy and corrupts correspondence; typical frame rates undersample the refractive field's coherence time, making fusion unstable with ghosts and limiting structural recovery and temporal consistency.

To mitigate these issues, we introduce event cameras [20], [25], [26], [27], [28]. Events asynchronously record per-pixel brightness changes with microsecond resolution and ultra-high dynamic range; their sampling typically exceeds the turbulence coherence time, preserving edge trajectories that frame integration smooths out, while HDR reduces scintillation-driven contrast fluctuations. Precise timestamps provide edge-centric anchors for estimating spatially varying deformations, and event density and polarity furnish directional and confidence cues that localize reliable structure and suppress strongly disturbed regions. Overall, events complement RGB in temporal precision and illumination robustness, supplying stable structural information for alignment and fusion under time-varying, anisoplanatic distortions.

In this letter, we propose a novel turbulence restoration framework, called Event-guided Turbulence Mitigation (EvTurM), which systematically integrates event data for restoring turbulence-degraded images. The proposed EvTurM consists of two core modules: (1) an *Event-Aware Modality Enhancement* (EAME) module that exploits dynamic cues from event streams to adaptively enhance spatial representations in RGB features; and (2) a *Bidirectional Modality Calibration* (BMC) module that strengthens temporal modeling across frames by fusing cross-modal features in both forward and backward directions. Together, these modules fully utilize the spatial-temporal complementarity between RGB and event data, leading to more accurate, stable, and consistent restoration under atmospheric turbulence.

The contributions of this work are summarized as follows: (1) We present EvTurM, an event guided framework for atmospheric turbulence restoration that fuses events with RGB to address space variant warps and temporal jitter. (2) We design a pipeline with two modules: the EAME module, which uses event dynamics to refine spatial representations, and the BMC module, which propagates cross modal cues forward and backward to improve temporal consistency and information flow. (3) Extensive experiments across turbulence levels and datasets show consistent gains in image quality and temporal stability over leading single-frame and multi-frame baselines.

II. METHOD

A. Overview

EvTurM is a practical framework for atmospheric turbulence mitigation that introduces event data into the task to assist image reconstruction. Given a sequence of turbulence-degraded image frames $\mathcal{I}^n = \{\dots, I_{t-1}^n, I_t^n, I_{t+1}^n, \dots\}$ ($\mathcal{I}^n \in$

Received 15 September 2025; revised 19 November 2025; accepted 22 December 2025. Date of publication 26 December 2025; date of current version 23 January 2026. This work was supported by the National Natural Science Foundation of China under Grant 62272142 and Grant 62476077. The associate editor coordinating the review of this article and approving it for publication was Dr. Seisuke Kyochi. (*Corresponding author: Zeyu Xiao.*)

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Code will be available at: GitHub: <https://github.com/yoong670/EvTurM>.

This article has supplementary downloadable material available at <https://doi.org/10.1109/LSP.2025.3648967>, provided by the authors.

Digital Object Identifier 10.1109/LSP.2025.3648967

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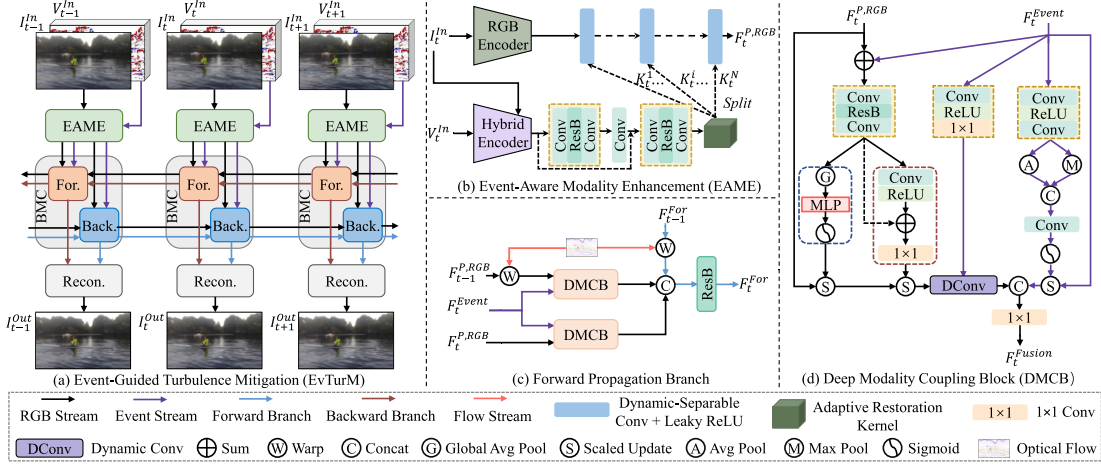


Fig. 1. Overview of the proposed EvTurM and its key modules. (a) Given RGB frames and the event stream, the EAME module enhances per-frame RGB features; forward/backward propagation branches in the bidirectional modality calibration (BMC) module aggregate cross-frame evidence, and the reconstructor outputs the de-turbulent frames. (b) Structure of the EAME module. (c) Structure of the forward propagation branch. (d) Structure of the DMCB block.

$\mathbb{R}^{T \times H \times W \times 3}$) and the corresponding event streams $\mathcal{E}^{In} = \{\dots, E_{t-1}^{In}, E_t^{In}, E_{t+1}^{In}, \dots\}$ triggered within T , E_t^{In} denotes the event stream at the time stamp t . EvTurM is designed to reconstruct image frames $\mathcal{I}^{Out} = \{\dots, I_{t-1}^{Out}, I_t^{Out}, I_{t+1}^{Out}, \dots\}$ ($\mathcal{I}^{Out} \in \mathbb{R}^{T \times H \times W \times 3}$) degraded by atmospheric turbulence by effectively leveraging event-based information, thereby enhancing spatial details and improving temporal consistency. $\mathcal{I}^{Out}$ should be close to the ground truth $\mathcal{I}^{GT} = \{\dots, I_{t-1}^{GT}, I_t^{GT}, I_{t+1}^{GT}, \dots\}$ ($\mathcal{I}^{GT} \in \mathbb{R}^{T \times H \times W \times 3}$). T , H , and W denote the number of frames, height, and width of the image, respectively. Since event data are sparse, irregular, and unstructured and thus unsuitable for convolutional networks [14], [28], we voxelize $\mathcal{E}^{In}$ into the event grid $\mathcal{V}^{In} \in \mathbb{R}^{T \times H \times W \times B}$ for processing (see supplementary document for more details).

Figure 1(a) illustrates the overall architecture of the proposed EvTurM. EvTurM adopts BMC consisting of a forward propagation unit F_f and a backward propagation unit F_b , similar to the design presented in [1], [25]. However, EvTurM incorporates event information as a key modality and integrates dedicated feature extraction and fusion modules to fully exploit the dynamic cues embedded in the event stream, thereby structurally distinguishing itself from previous RGB-only approaches.

At each time step t , the input RGB frame I_t^{In} and its corresponding event voxel V_t^{In} are jointly fed into the EAME module to alleviate dynamic information loss and enhance RGB feature representation, generating $F_t^{P,RGB}$ via $F_t^{P,RGB} = \text{EAME}(I_t^{In}, V_t^{In})$. Then, the event feature F_t^{Event} is extracted from the voxel input V_t^{In} using a lightweight convolutional encoder via $F_t^{Event} = \phi_{\text{conv}}(V_t^{In})$, where $\phi_{\text{conv}}(\cdot)$ is composed of a single 3×3 convolutional layer, a BatchNorm layer, and a ReLU activation function. Subsequently, both $F_t^{P,RGB}$ and F_t^{Event} are input into the BMC module for temporal modeling. The forward propagation unit F_f additionally receives the RGB feature $F_{t-1}^{P,RGB}$ and F_{t-1}^{For} from the previous time step, while the backward propagation unit F_b takes corresponding information from the next frame. This process can be denoted as:

$$\begin{aligned} F_t^{For} &= F_f(F_t^{P,RGB}, F_t^{Event}, F_{t-1}^{P,RGB}, F_{t-1}^{For}), \\ F_t^{Back} &= F_b(F_t^{P,RGB}, F_t^{Event}, F_{t+1}^{P,RGB}, F_{t+1}^{Back}). \end{aligned} \quad (1)$$

Finally, the reconstructor R fuses the forward feature F_t^{For} and backward feature F_t^{Back} and is applied in a residual manner with respect to the current frame I_t^{In} to produce the turbulence-mitigated result I_t^{Out} . The module comprises several convolutional layers that aggregate these inputs to recover structural details and enhance spatial clarity. Formally,

$$I_t^{Out} = I_t^{In} + R([F_t^{For}, F_t^{Back}]), \quad (2)$$

where $[\cdot, \cdot]$ denotes channel-wise concatenation.

B. Event-Aware Modality Enhancement

Turbulence causes space-variant blur and nonrigid warps that erase RGB details, whereas the event stream, with high temporal resolution and HDR, captures minute intensity changes within short intervals. To exploit this complementarity, we introduce EAME (Fig. 1(b)), which enhances RGB features with event-conditioned dynamic kernels and provides a more stable basis for subsequent restoration.

EAME comprises a primary and an auxiliary branch. The primary branch encodes the input RGB and applies a stack of depthwise separable convolutions (DS Convs) with activations. The kernels of these DS Convs are dynamically predicted by the auxiliary branch. The auxiliary branch concatenates RGB and events channel-wise, extracts fused features with a hybrid encoder, and passes them through a geometry estimator and a kernel generator to produce a high-dimensional kernel block, which is then split into layer-wise adaptive restoration kernels aligned with the DS Convs to correct turbulence-induced blur and geometric distortion:

$$\begin{aligned} F_t^{RGB} &= \phi_{rgb}(I_t^{In}), \\ \{K_t^1, \dots, K_t^n\} &= \text{Split}(\phi_{ker}(\phi_{geo}(\phi_{hyb}([I_t^{In}, V_t^{In}])))), \\ F_t^{P,RGB} &= \mathcal{F}(K_t^n, \dots, \mathcal{F}(K_t^2, \mathcal{F}(K_t^1, F_t^{RGB}))), \end{aligned} \quad (3)$$

where $\phi_{rgb}(\cdot)$ and $\phi_{hyb}(\cdot)$ are the RGB and hybrid encoders, $\phi_{geo}(\cdot)$ and $\phi_{ker}(\cdot)$ denote the geometry estimator and kernel generator, and $\mathcal{F}(K_t^i, \cdot)$ is the i -th DS Conv followed by LeakyReLU guided by kernel K_t^i .

By producing spatially and temporally adaptive kernels from both modalities, EAME sharpens true edges, recovers dynamic details, and strengthens cross-modal collaboration, improving

the structural expressiveness of the RGB path and stabilizing downstream turbulence mitigation.

C. Bidirectional Modality Calibration

To cope with severe spatial distortion and temporal inconsistency, BMC serves as the core spatio-temporal module in EvTurM. Following the bidirectional propagation idea of BasicVSR [1], BMC builds symmetric forward and backward units to transmit information across time [3], [4], [7], [10], [13], [17], [23], [34], [35], [36], [37], [38]. Unlike conventional designs [1], [2], [21], BMC jointly models RGB and events. Because the two modalities differ in sensing and temporal distribution, naïve coupling fails under strong spatio-temporal misalignment; we therefore introduce the DMCB to adaptively align and fuse features in the latent space, leveraging events for temporally precise dynamics while strengthening RGB representations for stable propagation and detail restoration.

a) *Flow-guided alignment*: We adopt a flow estimator S (e.g., SpyNet [18]) for short-term motion:

$$O_{t \rightarrow t-1} = S(I_t^{In}, I_{t-1}^{In}), O_{t \rightarrow t+1} = S(I_t^{In}, I_{t+1}^{In}). \quad (4)$$

Features from adjacent frames are warped to t using $\mathcal{W}(\cdot)$.

b) *Forward unit*: We align the previous predicted RGB feature and the forward state via:

$$\tilde{F}_{t-1}^{P,RGB} = \mathcal{W}(F_{t-1}^{P,RGB}, O_{t \rightarrow t-1}), \tilde{F}_{t-1}^{For} = \mathcal{W}(F_{t-1}^{For}, O_{t \rightarrow t-1}), \quad (5)$$

then fuse history and current cues with two DMCBs:

$$\begin{aligned} F_{t,his}^{Fusion} &= \phi_{DMCB}(\tilde{F}_{t-1}^{P,RGB}, F_t^{Event}), \\ F_{t,cur}^{Fusion} &= \phi_{DMCB}(F_t^{P,RGB}, F_t^{Event}), \\ F_t^{For} &= ResB([F_{t,his}^{Fusion}, F_{t,cur}^{Fusion}, \tilde{F}_{t-1}^{For}]). \end{aligned} \quad (6)$$

The backward unit is symmetric and propagates information from future to current using $O_{t \rightarrow t+1}$, yielding F_t^{Back} .

c) *Deep modality coupling block*: Given RGB and event features at time t , $F_t^{P,RGB}$ and F_t^{Event} , DMCB first extracts deep cross-modal cues:

$$F_t^{fused} = Conv(ResB(Conv(F_t^{P,RGB} + F_t^{Event}))). \quad (7)$$

Channel attention is derived from global average pooling and an MLP with a sigmoid, and applied via a scaled update to the RGB path via:

$$A_t^{ch} = \sigma(MLP(GAP(F_t^{fused}))), \quad (8)$$

$$F_t^{CA,RGB} = F_t^{P,RGB} + A_t^{ch} \otimes F_t^{P,RGB}. \quad (9)$$

Spatial attention uses local responses of F_t^{fused} :

$$A_t^{sp} = Conv_{1 \times 1}(F_t^{fused} + ReLU(Conv(F_t^{fused}))), \quad (10)$$

$$F_t^{Att,RGB} = F_t^{CA,RGB} + A_t^{sp} \odot F_t^{CA,RGB}. \quad (11)$$

A position-aware dynamic kernel from events modulates the RGB feature:

$$K_t^{Event} = Conv_{1 \times 1}(ReLU(Conv(F_t^{Event}))), \quad (12)$$

$$F_t^{D,RGB} = D_{Conv}(F_t^{Att,RGB}, K_t^{Event}). \quad (13)$$

In parallel, an event spatial-attention branch enhances F_t^{Event} :

$$\begin{aligned} \hat{F}_t^{Event} &= Conv(ReLU(Conv(F_t^{Event}))), \\ A_t^{Event} &= \sigma \\ &\times (Conv([AvgPool(\hat{F}_t^{Event}), MaxPool(\hat{F}_t^{Event})])), \\ F_t^{Att,Event} &= F_t^{Event} + A_t^{Event} \odot F_t^{Event}. \end{aligned} \quad (14)$$

Finally, RGB and event streams are fused via:

$$F_t^{Fusion} = Conv_{1 \times 1}([F_t^{D,RGB}, F_t^{Att,Event}]). \quad (15)$$

Here, $\otimes$ channel-wise multiplication, $\odot$ spatial-wise multiplication, and σ the sigmoid. DMCB thereby performs adaptive alignment and deep fusion, stabilizing cross-frame propagation under turbulence while retaining event-driven dynamics.

III. EXPERIMENTS

A. Experimental Settings

a) *Datasets*: We evaluate on ATSyn-dynamic [32], a synthetic video dataset for dynamic turbulence restoration. Some sequences with negligible motion fail to produce valid event streams under the RIFE+ESIM pipeline and are therefore removed, resulting in a slightly smaller but more challenging subset that is used for all methods. RGB videos (30 FPS) are temporally upsampled using RIFE [5] with an adaptive interpolation rate based on motion intensity, providing denser sampling during strong dynamics. Corresponding event streams are then generated with ESIM [19]. The final benchmark contains 3,819 training and 967 testing videos, covering diverse spatio-temporal turbulence patterns, after filtering out samples that cannot yield meaningful event streams.

b) *Training settings*: During training, we use the original-resolution ATSyn-dynamic images [32] as input and learn a direct mapping from turbulence-degraded to clean images. The objective minimizes the Charbonnier loss [24] between the output and the ground truth via $\mathcal{L} = \sqrt{(I^{GT} - I^{Out})^2 + \varepsilon^2}$, with $\varepsilon = 1 \times 10^{-6}$. A pre-trained SpyNet [18] estimates optical flow within the bidirectional units; its weights are frozen for the first 5,000 iterations,

allowing randomly initialized EAME/BMC to warm up on fixed flow features before jointly fine-tuning SpyNet [18]. We optimize with Adam ($\text{lr} = 1 \times 10^{-5}$, $\beta_1 = 0.9$, $\beta_2 = 0.99$) and cosine annealing to a minimum lr of 1×10^{-7} . The input patch size is 64×64 , the mini-batch size is 5, and training runs for 300 K iterations on five NVIDIA vGPUs.

c) *Evaluation metrics*: We evaluate using PSNR and SSIM, computed on RGB channels without border cropping.

B. Quantitative and Qualitative Comparisons

We compare EvTurM with representative multi-frame turbulence mitigation methods: TMT [33], VRT [11], RNN-MBP [40], ESTRNN [39], RVRT [12], and DATUM [32]. For fairness, all models are retrained on the same ATSyn-dynamic dataset [32] using their public implementations, without pre-trained weights or additional data.

a) *Quantitative results*: As summarized in Table 1, EvTurM outperforms all competitors across turbulence levels, especially versus DATUM. EvTurM yields consistent gains: weak +0.54 dB PSNR and +0.0050 SSIM; medium +0.42 dB and +0.0031 SSIM; strong +0.71 dB and +0.0045 SSIM; overall +0.55 dB and +0.0043 SSIM. These results indicate that high temporal resolution event cues improve both restoration fidelity and temporal modeling.

b) *Qualitative results*: Figure 2 compares EvTurM with representative RGB-only methods on ATSyn-dynamic. EvTurM recovers sharper edges and finer textures in the yellow zooms (e.g., sign/box and runner contours) and suppresses smear/ghosting from misalignment, producing images closer to the ground

TABLE I
QUANTITATIVE COMPARISON FOR ATMOSPHERIC TURBULENCE MITIGATION ON THE ATSyn-DYNAMIC DATASET IN TERMS OF PSNR AND SSIM. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Turbulence Level	Weak		Medium		Strong		Overall		Cost	
Methods	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	#Params	FPS
TMT [33]	28.10	0.8568	27.03	0.8389	26.15	0.8196	27.17	0.8398	26.04	0.80
VRT [11]	27.62	0.8521	27.00	0.8359	25.70	0.8132	26.84	0.8351	18.32	0.17
RNN-MBP [40]	26.60	0.8415	26.08	0.8237	24.93	0.8085	25.93	0.8258	14.16	1.14
ESTRNN [39]	27.77	0.8491	27.12	0.8315	25.79	0.8037	26.96	0.82970	2.468	27.65
RVRT [12]	28.52	0.8582	27.47	0.8415	26.57	0.8176	27.60	0.8405	13.50	2.43
DATUM [32]	29.09	0.8592	28.32	0.8437	27.02	0.8219	28.22	0.8429	5.754	9.17
EvTurM (Ours)	29.63	0.8642	28.74	0.8468	27.73	0.8264	28.77	0.8472	7.451	7.56

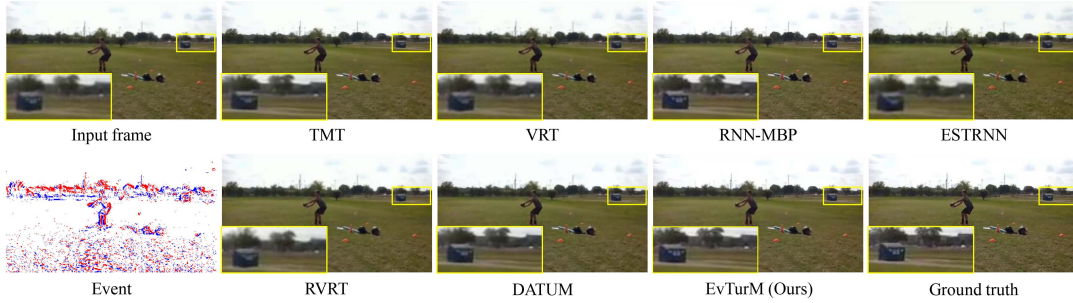


Fig. 2. Visual comparison between EvTurM and existing methods on the turbulence mitigation task. Samples are taken from the ATSyn-dynamic dataset.

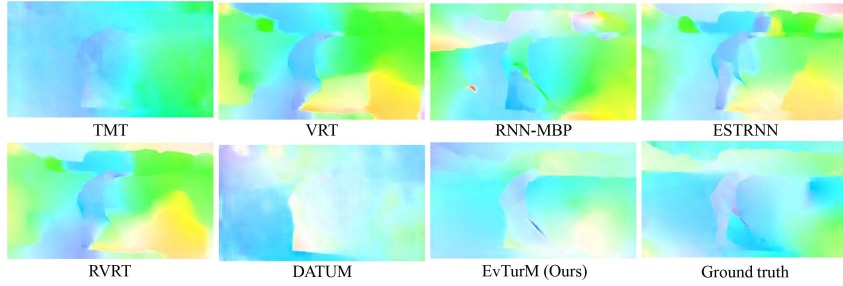


Fig. 3. Temporal consistency comparison in atmospheric turbulence mitigation. We visualize the estimated optical flow maps of restored sequences from different methods using the pre-trained RAFT [22], providing an intuitive assessment of their temporal modeling capabilities.

truth. By contrast, baseline methods tend to over-smooth high-frequency details.

c) Temporal consistency: To assess temporal consistency, we estimate adjacent-frame flow on ATSyn-dynamic with RAFT [22]. As shown in Fig. 3, flows from EvTurM reconstructions closely match the ground truth, whereas competing methods exhibit discontinuities or distortions in turbulent regions, confirming EvTurM's superior temporal coherence and stable restoration under dynamic degradation.

C. Ablation Studies

To evaluate the contributions of the two core modules, we ablate EvTurM on ATSyn-dynamic (Table 2). (1) EvTurM-w/o-EAME: removing EAME reduces performance to 28.32 dB/0.8402, indicating that event-aware enhancement accounts for a major gain. (2) EvTurM-w/o- F_b : disabling the backward branch yields 28.55 dB/0.8438, showing the benefit of future-context calibration. (3) EvTurM-w/o- F_f : disabling the forward branch gives 28.56 dB/0.8441; the similar drops in (2) and (3) indicate complementary roles of both directions. With both modules enabled, EvTurM reaches 28.77 dB/0.8472.

TABLE II
EFFECTIVENESS OF THE EAME MODULE AND THE BMC MODULE

Method	EAME	BMC		PSNR	SSIM
		F_b	F_f		
EvTurM-w/o-EAME	✗	✓	✓	28.32	0.8402
EvTurM-w/o- F_b	✓	✗	✓	28.55	0.8438
EvTurM-w/o- F_f	✓	✓	✗	28.56	0.8441
EvTurM	✓	✓	✓	28.77	0.8472

Further model analyses, including efficiency and deployment, are in the supplementary document.

IV. CONCLUSION

We presented EvTurM, an event-guided practical framework for atmospheric turbulence mitigation with two key modules: EAME, which injects event-conditioned dynamic kernels to refine RGB features, and BMC, which fuses RGB–event cues with forward and backward propagation branches. Experiments on ATSyn-dynamic show consistent gains in detail reconstruction and temporal consistency, with improved robustness, over strong RGB-only baselines.

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